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Dear SpaceX team,

I am a student at the University of Pennsylvania pursuing an MSE in Robotics and a BSE in Artificial Intelligence, and I'm excited to apply for the Spring 2026 Software Engineering Internship at SpaceX. I am drawn to SpaceX's mission to make life multiplanetary and to the opportunity to work on software that must be reliable, performant, and production-ready. My background building full-stack systems, machine learning pipelines, and physical systems makes me well suited to contribute to the Starship Software team.

Two accomplishments that best illustrate how I will add value at SpaceX are: (1) my work as an Autonomous Driving Research Intern, where I led data-pipeline and model-training work that produced a first-author publication and reduced training time on large models from 36 GPU hours to 24 GPU hours while preserving accuracy; and (2) co-founding and co-captaining my high school's Solar Car team, where I led a nine-person design and manufacturing effort (raising \$6,500+, securing 17+ sponsors), CAD-modeled and validated chassis components in Fusion 360, and helped the team place 4th nationally. Those experiences show my ability to ship technically sophisticated software and systems, optimize performance under tight constraints and short deadlines, and lead multidisciplinary teams through hands-on engineering work.

Technically, I contribute value across the stack. I've built full-stack AI systems (FastAPI backends + GPT-based assistants, LangChain integrations, PostgreSQL row-level security), authored many Python projects and PyTorch models (including RL for drone control), and used React/JavaScript for front-end UIs. I'm comfortable with debugging, performance optimization, unit testing, and producing clear system documentation and diagrams. I learn quickly, collaborate effectively, and thrive in environments where requirements evolve rapidly.

I am available for a full-time, on-site internship for the 12+ consecutive week period beginning January 2026 or March 2026, and I would welcome the chance to discuss how my technical background and hands-on leadership can support your software teams.

Sincerely,
Pranav Gupta